

Bobby Tromm

Education

- 2025–Present **PhD, Computational Neuroscience**, *University College London*, London, UK
Supervisors: Caswell Barry, Tim Behrens
Thesis: Cognitive maps in biological and artificial neural networks
- 2021–2023 **MSc, Cognitive and Clinical Neuroscience**, *Maastricht University*, Maastricht, Netherlands
- 2016–2020 **BS, Neuroscience**, *Brandeis University*, Waltham, USA

Research Experience

- 2024–2025 **Research Engineer**, *Neuronal Circuits & Brain Dynamics, Paris Brain Institute*, Paris, France
Supervisor: Nikolas Karalis
- 2023 **Visiting Researcher**, *Department of Psychiatry, University of Oxford*, Oxford, UK
Supervisor: Morten Kringelbach
- 2019–2020 **Research Assistant**, *Volen National Center for Complex Systems, Brandeis University*, Boston, USA
Supervisor: Eve Marder

Publications

- 2025 C. Coleman, K. Shinozuka, **R. Tromm** et al. *The role of the dorsolateral prefrontal cortex in ego dissolution and emotional arousal during the psychedelic state*. Human Brain Mapping. [\[Link\]](#)

Conference Presentations

- 2024 **R. Tromm**, C. Timmermann, P. Mallaroni, N. Mason, R.L. Carhart-Harris, G. Deco, J. Ramaekers, and M. Kringelbach. *Brain hierarchy under psychoactive substances*. ASSC 27. Tokyo, Japan. (Poster)
- 2023 **R. Tromm**, C. Timmermann, P. Mallaroni, N. Mason, R.L. Carhart-Harris, G. Deco, J. Ramaekers, and M. Kringelbach. *The anarchically organized brain: changes in functional hierarchy in altered states*. ALPS 2023. University of Geneva, Geneva, Switzerland. (Poster)
- 2019 **R. Tromm**, C. Guerini, S. Gorur-Shandilya, E. Marder. *Dual homeostatic mechanisms can reproduce diverse ion channel correlations*. SciFest IX. Brandeis University, Waltham, United States. (Poster)

Teaching & Supervision

- 2026–Present **BSc Project Supervisor**, *University College London*
Necessary architectures for grid cells
- 2025–Present **Postgraduate Teaching Assistant**, *University College London*
- COMP0085 Approximate Inference and Learning in Probabilistic Models (2026)
 - COMP0086 Probabilistic and Unsupervised Learning (2026)
 - NEUR0016 Neural Computation (2025)

Selected Honors & Awards

- 2025 **Research Excellence Scholarship**, *University College London*
£175,000, awarded to the top 0.5% of PhD applicants
- 2024 **Polaris Fellowship**, *Entrepreneur First*
Fellowship for technical researchers pursuing commercial research projects; £3,000 grant

- 2024 **Usona Institute Scholarship**, *Usona Institute*
Awarded twice (2023, 2024), \$7,500 each, for academic and research achievements in psychedelic studies
- 2020 **Justice Brandeis Scholarship**, *Brandeis University*
\$140,000, offered to top applicants

Activities & Societies

- 2026–Present **Co-organiser, AI Workshop 2026**, *Mediterranean Society for Consciousness Science*
Convening a workshop bridging consciousness science and mechanistic interpretability
- 2026–Present **As1 Next Generation Advisory Council**, *Burroughs Wellcome Fund*
- 2024–Present **Student Committee**, *Association for the Scientific Study of Consciousness*
- 2019–2021 **Research and Professional Development Committee**, *Intercollegiate Psychedelics Network*

Additional Training

- Aug 2026 **School on Analytical Connectionism**, *Chalmers University of Technology*
Statistical physics of learning and mechanistic interpretability of language acquisition
- Jul 2026 **Deep Learning**, *Neuromatch Academy*
Selective summer school in deep learning
- Jun 2025 **Gatsby Bridging Programme**, *Gatsby Computational Neuroscience Unit, UCL*
Mathematics summer school: linear algebra, calculus, probability, differential equations, Fourier analysis
- Aug 2024 **Neurohackademy**, *University of Washington*
Summer school in neuroimaging and data science
- Jun 2024 **QStr Summer School**, *ATR; Monash University*
Summer school on category theory and information theory in cognitive science
- Jun 2020 **Computational Neuroscience**, *Neuromatch Academy*
Single-neuron, dynamical systems, and stochastic modelling